Aliah University

Department of Electronics & Communication Engineering

(B.Tech 4th sem, ECE)

Paper Name: Computer Organization

Paper Code: CSEUGOE02

Time: 3hours **F.M: 80**

Group-A

(Multiple Choice Type Questions)

Answer the following questions. **1X10=10**

pages of memory

1.

i. What does Frame mean in Computer Organization?

ii. Describe the term "Instruction Cycle"?

iii. What do you mean by Associative Memory?

iv. Which addressing mode is used in instruction "POP B".

v. Write the function of the Program Counter.

vi. Write the value of (10101) after the *Arithmetic Shift Left* of the *Register content twice.*

vii. Represent the no. (-25) in 2's complement (8-bit format).

viii. What does 'Spatial locality of Reference' mean?

ix. What do you mean by Virtual memory?

x. Explain the concept of DMA.

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Group-B **5X6=30**

(Short Answer Type Questions)

Answer any **six** questions.

Address = 500	301
Next Instruction	302
550	499
800	500
900	599
700	600
450	800

2. Explain Index-based Addressing with proper examples. If the value of the *Register is 600*, then find the effective address and accumulator content for *Direct, Indirect, Immediate and Register Indirect* addressing mode. (1+4)

3. Apply Booth's algorithm to multiply the two numbers (-7) and (5). Assume the multiplicand and multiplier to be 4 bits each. 5

4. A cache has a 64KB capacity with Block size 128B and 4-way set associative. The CPU generates a 32-bit address for accessing data in the memory. How many bits are required to represent the Block offset, Set no. and Tag field? 5

5. Write down the functions of a Control Unit? Explain IEEE-754 standard format for floating point number representation in single precision and double precision format. (2+3)

6. Find the addition of (+0.75) & (0.125) using 8-bit representation. Mention the different *Fields of Instruction*. (3+2)

7. Discuss the Design and characteristics of the Memory Hierarchy. 5

8. What do you mean by Hit Ratio and Miss Ratio in Cache memory? If a processor's TLB hit ratio is 80% and it takes 20ns to search the TLB and 100ns to access main memory. What will be the effective access time? (2+3)

Group-C

10x4=40

(Long Answer Type Questions)

Answer any **four** questions.

9. What do you mean by pipeline hazards? Draw the space-time diagram for 4- the stage/segment pipeline and compare the time taken to complete the 5-instruction with non-pipeline architecture. In a pipeline architecture, the time delay for 4 stages τ_1=150ns, τ_2=140ns, τ_3=180ns, and τ_4=170ns, and their interface latch has a delay of 20ns, then find the Frequency, Clock Period, and Speed Up. (2+4+4)

10. Daisy chaining interrupt is which type of interrupt explain it. Discuss the concept of a reservation table for linear and non-linear pipelining architecture with diagram. Find Permissible latency and Collision Vector for the following reservation table. (2+5+3)

	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
S1	X_1		X_2		X_3			X_4	
S2		X_1			X_2		X_3		X_4
S3			X_1			X_2			X_3

11. Explain various address mapping techniques of the **Cache Memory**. 10

12. Define the term Access time, Seek time, and Latency in case of Magnetic disk accessing. Consider a 4-way set associative cache with 16 cache lines. The Main Memory block requirements are 0,255,2,4,3,8,133,159,26,129,63,8,48,32,73,92,155. Assumes initially the cache is empty. Which blocks are currently in Cache Memory using a)**FIFO** and b)**LRU** separately. (3+7)

13. Write short notes in any *two* of the following: 5+5

    i. Addressing modes
    ii. Associative Memory
    iii. RAID

Data
Instruction

B.Tech Examination-2023
Electronics and Communication Engineering
(Even Semester Regular and Supplementary)
Course Name: Electromagnetic Engineering, Course Code: ECEUGPC06

Full Marks: 80 **Time: 3.00 Hrs**

- **Answer Q.1 and seven questions from rest of the questions.**
- **Answer all parts of a question in same place.**
- **Symbols have their usual meaning**

1. **Choosing the correct alternative from the given options answer any ten questions.** 10x1=10
   - (a) Which of the following is a scalar quantity?
     (i) Electric displacement density (ii) Potential in electric field (iii) Electric field strength (iv) None of the above
   - (b) The force between two magnetic poles varies with the distance between them. The variation is ________ to the square of that distance.
     (i) equal (ii) greater than (iii) directly proportional (iv) inversely proportional
   - (c) Displacement current is due to
     (i) displacement of electric charge (ii) time varying magnetic field (iii) time varying electric field (iv) both (ii) and (iii)
   - (d) Equation of continuity is based on
     (i) conservation of charge (ii) conservation of momentum (iii) conservation of angular momentum (iv) none of these
   - (e) Poynting theorem represents
     (i) conservation of charge (ii) conservation of momentum (iii) conservation of energy (iv) none of this
   - (f) According to Maxwell's equation in free space $\nabla.\mathbf{E} = ?$
     (i) ρ (ii) $\frac{\rho}{\varepsilon_0}$ (iii) 0 (iv) $\frac{\varepsilon_0}{\rho}$
   - (g) Maxwell observed and corrected a discrepancy in
     (i) Ampere's Law (ii) Faraday's Law (iii) Gauss's Law for Electromagnetic (iv) Snell's law
   - (h) In a conducting medium, the electromagnetic waves are
     (i) amplified (ii) attenuated (iii) both (i) & (ii) (iv) none of these
   - (i) The wave velocity in non-conducting medium is
     (a) $\frac{1}{\sqrt{\mu\varepsilon}}$ (b) $\sqrt{\frac{\mu}{\varepsilon}}$ (c) $\frac{\mu_0}{\varepsilon_0}$ (d) $\frac{1}{\sqrt{\mu_0\varepsilon_0}}$
   - (j) A transmission line is said to be distortionless line when
     (i) $\frac{R}{L}=\frac{G}{C}$ (ii) $\frac{R}{G}=\frac{C}{L}$ (iii) $RG=\frac{L}{C}$ (iv) $\frac{R}{G}=LC$
   - (k) Which of the following is not a Maxwell's equation?
     (i) $\mathbf{D}=\varepsilon\mathbf{E}$ (ii) $\nabla.\mathbf{D}=\rho$ (iii) $\nabla\times\mathbf{E}=-\frac{\partial \mathbf{B}}{\partial t}$ (iv) $\nabla\times\mathbf{H}=\mathbf{J}+\frac{\partial \mathbf{D}}{\partial t}$
   - (l) The direction of propagation of electromagnetic wave is obtained from – (i) $\mathbf{E}\times\mathbf{H}$ (ii) $\mathbf{E}-\mathbf{H}$ (iii) $\mathbf{E}$ (iv) $\frac{\mathbf{E}}{\mathbf{H}}$

2. (a) State Divergence theorem and Stoke's theorem. 4
   (b) Show that the vector $H = (y^2 - z^2 + 3yz - 2x)\,\mathbf{a}_x + (3xz + 2xy)\,\mathbf{a}_y + (3xy - 2xz + 2z)\,\mathbf{a}_z$ is both irrotational and solenoidal. 3
   (c) A volume charge density is expressed as $\rho_v = 10z^2 e^{-0.1x} \sin \pi y$. Find the total charge inside the volume where $-1 \leq x \leq 2,\ 0 \leq y \leq 1,\ 3 \leq z \leq 3.6$ 3

3. (a) What do you mean by electric field intensity at a point? 2
   (b) Define electric flux and flux density. 2
   (c) Develop an expression for electric field intensity at a general point P due to an infinite straight line charge with charge density ρ_l C/m. 6

4. (a) State and prove the Gauss's Law. 4
   (b) Derive Poisson's and Laplace's equation. 3
   (c) The electric flux density is given as $\mathbf{D} = \frac{r}{4}\,\mathbf{a}_r$ nC/m^2 in free space. Calculate: (i) the electric field intensity at $r = 0.25$ m (ii) the total charge within a sphere of $r = 0.25$ m (iii) the total flux leaving the sphere $r = 0.35$ m. 3

5. (a) Derive the equation of continuity $\frac{\partial \rho_v}{\partial t} + \nabla.\mathbf{J} = 0$ 3
   (b) Prove that the tangential component of electric field intensity is continuous across the boundary between two perfect dielectric medium. 4

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P 6 5/30 FQ A2 9: FQ A

(c) In a certain region, $\mathbf{J} = (4y\mathbf{a}_x + 2xz\mathbf{a}_y + z^3\mathbf{a}_z)\sin(10^4 t)$ A/m; if volume charge density ρ_v in z=0 plane is 0, then find ρ_v.

$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$

6. (a) State Ampere's circuital law.
(b) A circular loop of wire of radius 'a' lying in xy plane with its centre at the origin carries a current **I** in the + φ direction. Using Biot-Savart law find **H** (0, 0, z) and **H** (0, 0, 0).
(c) The flux density at a point distance r from a long filamentary conductor carrying a current I in $\mathbf{a_z}$ direction is given by $\mathbf{B} = \frac{\mu_0 I}{2\pi r}\mathbf{a}_\varphi$. Calculate the flux crossing the portion of plane $\varphi = \frac{\pi}{4}$ defined by $0.01 < r < 0.05$ and $0 < z < 2$ for a current of 2A .

7. (a) State Faraday's law of electromagnetic induction.
(b) Prove that $\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$. The symbols have usual meaning.
(c) State two differences between conduction current density and displacement current density.
(d) Find the conduction and displacement current densities in material having conductivity of 10^{-3}S/m and ε_r=2.5 if electric field in the material is
$E = 5.0 \times 10^{-6}\sin(9.0 \times 10^9 t)$ V/m

8. (a) What is uniform plane wave? 2
(b) Show that for a uniform plane waves in a perfect dielectric medium, E and H are normal to each other and ratio of their magnitude is constant of the medium. What is the name of this constant? 2
(c) An electric field is represented by $\mathbf{E}_y = 10\cos(6\pi \times 10^8 t - \beta x)\mathbf{a}_y$ is propagating through a lossless medium having μ_r =1 and ε_r=78 at a frequency of 300 MHz. Find β, v, η and the corresponding magnetic field **H**. 4

9. (a) What do you understand by skin depth? 4
(b) Derive the expression for reflection and transmission coefficients when uniform plane wave incidents normally on the interface between two perfect dielectric medium and also find their relation. 2
(c) A large copper conductor with $\sigma = 5.8x10^7$ S/m and $\varepsilon_r = \mu_r = 1$ supports a uniform plane wave at 60 Hz. Determine the attenuation constant, phase constant and propagation constant. 5

10. (a) What are the types of transmission line? 3
(b) Starting from the equivalent circuit derive the transmission line equations for V and I, in terms of source parameter. 1
(c) Write the condition of distortion-less transmission line and using that condition find the expression of propagation constant, phase velocity and characteristics impedance. 5

11. (a) Find the expression for the input impedance Z_{in} of a short-circuited, an open-circuited and a matched lossless transmission line using standard expression of input impedance of transmission line. 4
(b) Define the term VSWR and reflection coefficient for transmission line and write the relation between them 3
(c) A 100 m long lossless transmission line has a total inductance and capacitance of 100 μH and 10 nF, respectively. Determine the velocity of propagation, phase constant and characteristic impedance of the transmission line at operating frequency of 100KHz. 4

3

-End-

$\boxed{E = -\frac{d\phi}{dt}}$

Spring (Even) Semester Examination, 2023

Subject Name: **Environmental Science** Subject Code: **UCCUGMC02**

Full Marks: 80 Time: 3:00 Hours

Group – A (Attempt all questions) **40 x 1 = 40**

Choose the correct answer.

1. The upper layer of the lithosphere is known as
   a. Stratosphere b. Crust c. Core d. Mantle

2. The planet Jupiter is a
   a. Gaseous planet b. Terrestrial planet c. Both gaseous and terrestrial d. None of these

3. One non-polluting energy resource is
   a. Coal b. Solar energy c. Fuel wood d. Petrolium

4. CFC is-
   a) Chlorofluoro Carbon b) Centre Fuel Control
   C) Carcinogenic Fluride Compound d) None of these

5. The most famous mass extinction happened in Cretaceous period when
   a. Dinosaur extinct b. Dodo bird extinct c. Passenger pigeon extinct d. none of this

6. The Bhopal gas disaster happened in India year
   a. 1864 b. 1964 c. 1984 d. 2012

7. Example of one keystone species is
   a. Cow b. rose plant c. Palm tree d. Banyan tree

8. The full form of MoEFCC is
   a. Ministry of environmental conference
   b. Member of environmental and forest conference
   c. Ministry of Environment, Forest and Climate change
   d. Ministry of Man and Climate change

9. One air pollutant reducing equipment is
   a. BOD bottle b. Electrostatic precipitator c. pH paper d. none of this

10. Lichen species is the indicator of
    a. acid rain b. dust particle c. radioactive substances d. global warming

11. The book "Origin of species" written by
    a. Rachel Carson b. Charles Darwin c. Ernst Haeckel d. Rabindranath Tagore

12. Silicosis is a
    a. waterborne disease b. airborne disease c. vector borne disease d. None of this

13. "The London Smog" incident happened in the year
    a.1947 b.1952 c. 2000 d. 2022

14. Thebiomass pyramid is
    a. always upright b. always inverted c. Upright or inverted d. none of this

15. Photochemical smog is the example of
    a. primary air pollutant b. secondary air pollutant
    c. biodegradable air pollutant d. none of these

16. One example of indoor air pollutant is
    a. cigarette smoke
    b. Dust particles from construction site
    c. Sulphur Dioxide from thermal power plants
    d. Smokes from automobiles

17. The percentage of Nitrogen in the atmosphere is
    a. 21% b. 52% c. 34% d. 78%

18. The Silent Valley movements started to protect the
    a. forest and biodiversity
    b. Kunti river
    c. Kerala region
    d. Atmosphere of the Silent Valley

19. Minamata disease occurred due to presence of — in water
    a. Sulphur dioxide
    b. Bacteria
    c. Mercury
    d. potassium cyanide

20. The World Environment Day is celebrated on
    a. First April
    b. 5th May
    c. 15th July
    d. 5th June

21. The Wildlife Protection Act come to force in the year
    a. 1942 b. 1972 c. 2014 d. 2022

22. The age of our Earth is
    a. 4 billion years b. 5 billion years c. 6 billion years d. 7 billion years

23. The number of National Park present in India (as on January, 2023) are-
    a.106 b. 206 c. 500 d. 1000

24. "Meeting the needs of present without compromising the ability of future generation to meet their own need" is the definition of sustainable development which was given by:
    a. Mahatma Gandhi
    b. G.H. Brundtland
    c. Wangari Maathai
    d. SunderlalBahuguna

25. The maximum number of individuals that can be supported by a given environment is called
    a. Carrying capacity
    b. Population size
    c. Biotic potential
    d. Environmental resistance

26. Social, economic and ecological equity is the necessary condition for achieving
    a.Ecological development
    b. Economic development
    c. Sustainable development
    d. Social development

27. The presence of high coliform counts in water indicate
    a. Contamination by human wastes
    b. phosphorous contamination
    c. Decreased biological oxygen demand
    d. Hydrocarbon contamination

28. The ecological pyramid that is always upright
    a. Pyramid of number
    b. Pyramid of biomass
    c. Both 'a' and 'b'
    d. Pyramid of energy

29. A high BOD value in aquatic environment is indicative of
    a. A pollution free system
    b. A highly polluted system due to excess of nutrients
    c. A highly polluted system due to abundant heterotrophs
    d. A highly pure water with abundant of autotrophs

30. World wetland day is celebrated on
    a. 2nd February b. 21st March c. 5th June d. 16th September

31. Which one is true?

a. Symbiosis is when neither population affects each other
b. Symbiosis is when the interaction is useful to both the population
c. Commensalism is when none of the interacting populations affect each other
d. Commensalism is when the interaction is useful to both the populations

32. Which of the following sources of energy do not produce carbon dioxide?

a. Wind energy
b. Geothermal energy
c. Hydroelectric energy
d. All of the above

33. Which one of the following is a correct grazing food chain?

a. Grass → Grasshopper → Frog → Snake → Hawk
b. Grass → Frog → Grasshopper → Snake → Hawk
c. Grass → Grasshopper → Frog → Hawk → Snake
d. Grass → Grasshopper → Snake → Frog → Hawk

34. Which one is sedimentary cycle?

a. Hydrogen cycle
b. Phosphorous cycle
c. Oxygen cycle
d. Nitrogen cycle

35. The Earth receive major energy from

a. Sun b. Moon c. Mars d. Jupiter

36. Non- living components of ecosystem are

a. Biotic b. Abiotic c. Free living d. None

37. Man belongs to

a. Herbovours b. Carniovours c. Omnivours d. none of these

38. Which of the below is producer

a. Cat b. Fish c. Tiger d. Paddy

39. Which of the following is a part of pond ecosystem?

a. Fish b. Lizard c. Goat d. Tiger

40. Mangrove type forest are found in-

a. Nyle Delta
b. Godabari delta
c. Mississippi delta
d. Sundarban delta

Group- B Attempt any FOUR questions from the following

1. Which layer is called 'tropopause'? Discuss about the layers of crust to inner core of our earth. 2+8= 10

2. What is biodiversity? Write the uses and values of biodiversity. What do you understand by the term 'hotspot'? 2+5+3= 10

3. Write short notes on the following; 5x2=10
   a. Conservation of biodiversity.
   b. Acid rain.
   c. Photochemical smog.
   d. Origin of oxygen in the primitive atmosphere.

4. Give a deliberate discussion on the 'Chipko Movement'. 10

5. What do you mean by pollution? On the basis of persistence, discuss about the types of air pollutants present in the atmosphere. Write the health impacts of air pollution. 2+4+4= 10

6. How many types of pyramid are present in the ecosystem? Discuss different types of ecological pyramids present in the ecosystem. 2 + 8= 10

Biology for Engineers

2023

Marks-80 **Time-3Hrs**

A. Answer the following question with select the correct options. **1×20=20**

1. This is the largest phylum of Animal on the earth.
A. Mollusca B. Amphibia C. Arthropoda D. Aves

2. Where is DNA present in the eukaryotic cells?
A. Inside the nucleus B. With other cellular contents C. Inside the ribosomes D. Not present

3. Which of the following nitrogenous base is not present in DNA?
A. Thymine B. Adenine C. Guanine D. Uracil

4. Whichofthefollowingisthefunctionoflysosomes?
A. Autophagy B.Autolysis C.Digestion D.All of the above

5. Who is known as the "Father of Genetics"?
A. Morgan B. Mendel C. Watson D. Bateson

6. Glycolysis is the conversion of
A. Fructose into phosphoenolpyruvate B. Fructose into pyruvate
C. Glucose into phosphoenolpyruvateD. Glucose into pyruvate

7. Which substrate is used in the last step of glycolysis?
A. Glyceraldehyde 3-phosphate B. Pyruvate
C. Phosphoenolpyruvate D. 1, 3-bisphosphoglycerate

8. Which of the following immunity is present from our birth?
A. Innate Immunity B. Active immunity C. Passive immunity D. Acquired immunity

9. Which of the following cells is involved in humoral immunity?
A. T-cells B. B-cells C. Mast cells D. Both T and B cells

10.Anticodon is present in
A. DNA B. tRNA C.tRNA D. mRNA

11. Name the RNA molecules which is used to carry genetic information copied from DNA?
A. tRNA B. mRNA C.RRNA D. snRNA

12. The percentage of human genome which encodes proteins is approximately
A. Less than 2% B. 5% C. 25% D. 99%

13. The jawless vertebrate is
A. Crocodile B. zoris C. *Hyla* D. *Petromyzon*

14. Phylum of *Taenia Solium* is
A. Aschelminthes B. Annelids C. platyhelminths D.Mollusca

15. On the Origin of Species was written by ______
A. Charles Darwin B. Ludmila Kuprianova C. Mikhail A. Fedonkin D. None of the above

16. Which of these is a characteristic of prokaryotic cells?

A. Absence of cell organelles B. Absence of nucleus
C. Presence of 70S ribosomes D. All of these

17. From which structure is a mesosome derived from?
A. Plasmid B. Cell wall C. Ribosome D. Cell membrane

18. Thescalelessvertebrateis
A. Snake B. Rohu C. Shark D. rat

19. Which of thefollowingisafightlessbird?
A. Pigeon B.vulture C.Parrot D. ostrich

20.Booklungsarerespiratory organsin
A. Insects B. Aarachnids C.Molluscans D. Echinoderms

B. Answer any 10 questions from the following. **2×10=20**

1. What is Coelom? Where is it found?
2. What is genetic code?
3. In which cell organelle thylakoid is found? Mention its one function.
4. Mention two differences betweenDNA and RNA.
5. Write down the two characteristic features of Phylum Arthropoda.
6. Mention the chemical reaction of photosynthesis?
7. What is Central Dogma in molecular biology?
8. Mention two differences betweenB-cell and T-cell .
9. What is mRNA? Mention its one function.
10. What is Green gland? Where it is found?
11. What do you mean by antibody?
12. What is chromosome?

C. Answer any 6 questions from the following. **5×6=30**

1. Write down the characteristics features of Phylum Mollusca. Mention two animal names under this phylum.
2. Write down a short note on Antibody structure.
3. Describe the Oparin and Haldane theory on the chemical basis of the origin of life.
4. What is nucleotide? Write down the structure of double helix DNA.
5. What is notochord? Write down the characteristics features of class Amphibia.
6. Write down the structure of mitochondria and mention its function.
7. What is immunity? Mention the different types of WBC. Mention its function in immunity.
8. What is metazoan? Write down thecharacteristics featuresof PhylumPoifera.

D. Answer any one question from the following **10×1=10**

1. What is Photosynthesis? Write down the process ofPhotosynthesis. Mention its importance. **(2+6+2)**

2. What is TCA cycle. Describe the Kreb cycle.Mention its importance. **(2+6+2)**

lycocytes
Monocytes
cistrae.
nucelic
nuclic
cistrae.

B.Tech. Examination-2023

Electronics and Communication Engineering Department

(Even Semester Regular and Supplementary)

Digital Electronics and Logic Design (ECEUGPC05) for ECE

Full Marks : 80 **Time-3 hrs**

(Answer Question no. 1 with proper reasoning and any four from the rest)

1. (a) In the circuit shown in Fig. 1, if C = 0, find the logic expression for Y. 3
   (b) For the logic circuit shown in Fig. 2, find the required input condition (A,B,C) to make the output X = 1. 3
   (c) Find the minimum number of 2-input NAND gates required to implement the Boolean function $Z = A\bar{B}C$, assuming that A, B and C are available. 3
   (d) Find the product of sums form of the Boolean function $F = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C}$. 3
   (e) The minimized form of the logic expression $\bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}BC + AB\bar{C}$ is 3
   (i) $\bar{A}\bar{C} + B\bar{C} + \bar{A}B$ (ii) $A\bar{C} + \bar{B}C + \bar{A}B$ (iii) $\bar{A}C + \bar{B}C + \bar{A}B$ (iv) $A\bar{C} + \bar{B}C + A\bar{B}$
   (f) The logic function implemented by the circuit shown in Fig. 3 is (ground implies a logic '0') 3
   (i) F = AND (P,Q) (ii) F = OR (P,Q) (iii) F = XNOR (P,Q) (iv) F = XOR (P,Q)
   (g) The circuit given in Fig. 4 is a 2
   (i) J-K flip-flop (ii) S-R flip-flop (iii) R-S latch (iv) D flip-flop

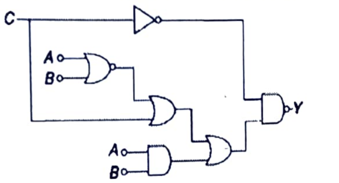

Fig. 1

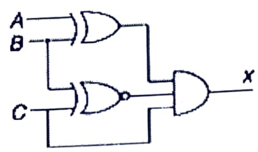

Fig. 2

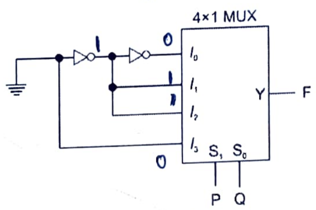

Fig. 3

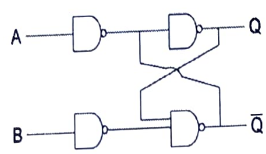

Fig. 4

xy + x̄z + xyz + x̄yz
x̄z(y+1) + xy + xyz
x̄z + xy(z+1)

2. (a) Prove the following theorems of Boolean algebra with algebraic methods only :- (3+4+4)
   (i) $x + yz = (x+y)(x+z)$ (ii) $xy + \bar{x}z + yz = xy + \bar{x}z$ (iii) $\overline{x_1 x_2} = \overline{x_1} + \overline{x_2}$
   (b) Show that $\overline{x \odot y} = x \oplus y$. 2
   (c) Show that NOR gate is an universal logic element. 2
3. (a) Explain the operation of 4-bit binary parallel adder/subtractor. 5
   (b) Construct a 5×32 decoder with four 3×8 decoders and a 2×4 decoder. 5
   (c) Implement a full-subtractor circuit with a 3 × 8 decoder and two OR gates. 5
4. (a) Simplify the following Boolean function in sum of products form using K-map method :- $F = \bar{B}\bar{C}\bar{D} + BC\bar{D} + ABC\bar{D}$; $d = \bar{B}C\bar{D} + \bar{A}B\bar{C}D$ 7
   (b) Obtain the simplified expression in product of sums form of the following logic function. $F(w,x,y,z) = \prod(1,3,5,7,13,15)$ 8
5. (a) Implement 8×1 MUX with two 4×1 MUX and OR gate. 7
   (b) Implement the following function with a multiplexer : $F(A,B,C,D) = \sum(0,1,3,4,8,11,15)$ 8
6. (a) Draw and explain the operation of mod-8 ripple counter with output timing waveforms. 2+4
   (b) Design a mod-3 counter with counting states 00, 10, 11, 00. 4+5
7. (a) Realize JK flip-flop from SR flip-flop and verify its truth table. 3+5
   (b) Convert JK flip-flop into D flip-flop and T flip-flop. 4+3
8. (a) With suitable logic diagram explain the operation of mod-4 ring counter. 3+6
   (b) Explain the operation of 4-bit shift register with timing waveform. 2+4
9. (a) Explain the operation of 4-bit R-2R ladder type D/A converter. 6
   (b) Draw the basic circuit of ROM cell and explain its working. 4
   (c) Explain with a circuit diagram the operation of TTL 3-input NAND gate. 5

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B.Tech Examination-2023
Dept. of Electronics and Communication Engineering
(Even Semester Regular & Supplementary)
Course Title: Electronic and Electrical Measurement
Course Code: ECEUGPC07

Full Marks: 80

Time: 3.00 Hrs

- ***Answer question <u>no. 1</u> and <u>any four</u> questions from the rest.***
- *Answer all parts of a question in same place.*
- *Figures on the right hand side margin indicate full marks.*
- *Symbols have their usual meaning*

Q. No.		Stem of the question	Marks
1	(a)	Define random errors and explain how they are analysed statistically.	2
	(b)	A $3\frac{1}{2}$ digit digital voltmeter has an internal reference of 200 mV. Calculate the full-scale count and resolution of the digital voltmeter.	2
	(c)	State the limitations of low resistance measurement using Wheatstone bridge.	2
	(d)	A 1 mA ammeter has a resistance of 100Ω it is to be converted to 1A ammeter. Calculate the value of shunt resistance.	2
	(e)	What is swamping resistance? For what purpose is swamping resistance used?	2
	(f)	Why the lower end of scale of an MI instrument is congested?	2
	(g)	What are Lissajous patterns?	2
	(h)	What are reasons for developing the residual voltage of LVDT?	2
2	(a)	What is a measuring Instrument? What is systematic error and how can we reduce it?	4
	(b)	A 0-150 V voltmeter has a guaranteed accuracy of 2% of full-scale reading. The voltage measured by the voltmeter is 75 volts. Determine the limiting error in percentage.	4
	(c)	With the help of a block diagram explain the various stages of a complete measuring instrument.	7
3	(a)	Explain the following terms associated with measuring instruments preferably with examples: (a) Accuracy, (b) Precision, (c) Resolution, (d) Sensitivity, and (e) Threshold.	5
	(b)	Drive the equation for deflection of a MI instrument if the instrument is spring controlled.	6
	(c)	A PMMC instrument has a coil of dimensions 15 mm × 12 mm. The flux density in the air gap is 1.8×10^{-3} wb/m^2 and the spring constant is 0.14×10^{-6} N-m/rad. Determine the number of turns required to produce an angular deflection of 90° when a current of 5mA is flowing through the coil.	5

4 (a) Describe the various operating forces needed for proper operation of an analog indicating instrument. 6

(b) Explain how eddy current damping is achieved in PMMC instrument. 4

(c) Draw and explain the operation of a Meggar used for high resistance measurement. 6

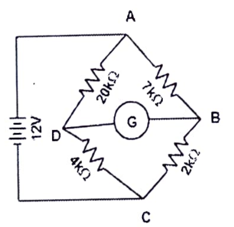
5 (a) Explain how Wien's bridge can be used for measurement of unknown frequency. Derive the expression for frequency in terms of bridge parameters and draw the phasor diagram. 10

(b) The four arms of a bridge are connected as follows: 6
Arm AB: A capacitor C_1 with an equivalent series resistance r_1,
Arm BC: A noninductive resistance R_3,
Arm CD: A noninductive resistance R_4,
Arm DA: A capacitor C_2 with an equivalent series resistance r2 in series with a resistance R_2.
A supply of 500 Hz is given between terminals A and C and the detector is connected between nodes B and D. At balance, $R_2=5\Omega$, $R_3=1000\Omega$, $R_4=3000\Omega$, $C_2=0.3\mu F$ and $r_2=0.25\Omega$. calculate the values of C_1 and r_1 and dissipation factor of the capacitor.

6 (a) What are the sources and detectors in ac bridges? 3

(b) Describe the working of schering bridge for measurement of capacitance and dissipation factor. Drive the equations for balance and draw the phasor diagram under balance condition. 8

(c) 5



Calculate the current flowing through the galvanometer shown in the above figure. The resistance of the galvanometer is 200Ω.

7 (a) What are the advantages of digital systems over analog? 3
(b) Explain the operating principle of integrating type DVM using a suitable block diagram. 7
(c) With the help of a functional block diagram, describe the principle of operation of a digital multimeter. 6

8 (a) Explain the working of a Digital Storage Oscilloscope with the help of a block diagram. 6

(b) Draw the block diagram of a Dual Trace Oscilloscope and explain the same. 6
(c) What is meant by deflection factor and deflection sensitivity of a CRO? 4

9 (a) Explain with the help of a functional block diagram, describe the principle of operation of a digital frequency meter. 6

(b) What is the difference between sensor and transducers? 3

(c) Draw the schematic diagram of an LVDT and explain its electro-mechanical transfer characteristics. 7